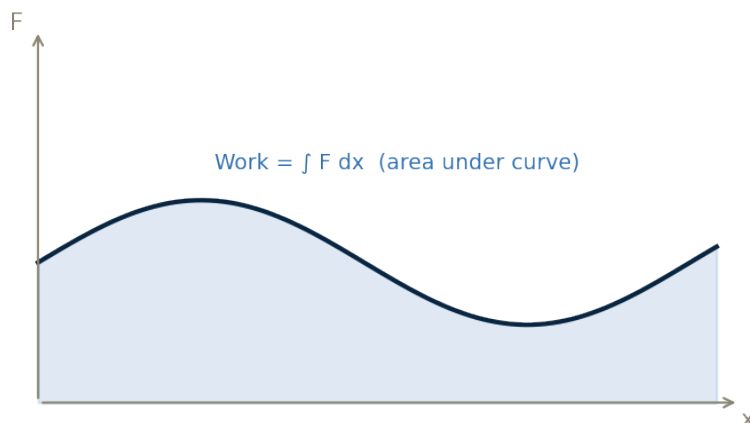


Work, Energy and Power

JEE Physics — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Work done by a constant force: $W = F \cdot s = F s \cos(\theta)$, where θ is the angle between force and displacement.
- Work-Energy Theorem: the net work done on a body equals its change in kinetic energy, $W(\text{net}) = \Delta KE$.
- Kinetic energy $KE = \frac{1}{2} m v^2$. Potential energy (gravitational) $U = mgh$; (spring) $U = \frac{1}{2} k x^2$.
- A force is conservative if the work done by it is independent of path (depends only on initial and final positions), e.g., gravity, spring force. Friction is non-conservative.
- Law of Conservation of Mechanical Energy: in the absence of non-conservative forces, $KE + PE = \text{constant}$.
- Power is the rate of doing work: $P = W/t = F \cdot v$ (instantaneous power = force dot velocity).
- Collisions: in a perfectly elastic collision, both momentum and kinetic energy are conserved. In a perfectly inelastic collision, momentum is conserved but kinetic energy is not (bodies stick together).



Force vs displacement graph: the area under the curve equals the work done.

2. Formula Sheet

Work done	$W = F s \cos(\theta)$
Kinetic energy	$KE = \frac{1}{2} m v^2$
Gravitational PE	$U = m g h$
Spring PE	$U = \frac{1}{2} k x^2$
Work-energy theorem	$W(\text{net}) = KE(\text{final}) - KE(\text{initial})$
Power	$P = W/t = F \cdot v$
Elastic collision velocity (1D, m1 hits stationary m2)	$v_1' = \frac{(m_1 - m_2)}{(m_1 + m_2)} u_1$; $v_2' = \frac{2m_1}{(m_1 + m_2)} u_1$
Perfectly inelastic collision	$v = \frac{(m_1 u_1 + m_2 u_2)}{(m_1 + m_2)}$

3. Practice Problems (22 questions)

1. A force of 20 N acting at 60 degrees to the horizontal displaces a body by 5 m along the horizontal. Find the work done.
2. A body of mass 2 kg falls freely from a height of 10 m. Find its kinetic energy just before hitting the ground ($g = 10 \text{ m/s}^2$).
3. A spring of spring constant 200 N/m is compressed by 0.1 m. Find the potential energy stored in it.
4. A block of mass 5 kg moving at 4 m/s is brought to rest by friction over a distance of 2 m. Find the work done by friction.
5. A machine does 5000 J of work in 20 s. Find its power output.
6. A car of mass 1000 kg accelerates from 0 to 20 m/s in 10 s. Find the average power delivered by the engine (neglect resistive forces).
7. A ball of mass 0.5 kg moving at 6 m/s collides with a stationary ball of mass 1.5 kg and they stick together. Find the common velocity after collision.
8. A 1 kg ball moving at 5 m/s collides elastically with a stationary 1 kg ball. Find the velocities of both balls after collision.
9. A body of mass 4 kg is lifted through a height of 3 m. Find the work done against gravity ($g = 10 \text{ m/s}^2$).
10. A pump lifts 200 kg of water through a height of 10 m in 5 s. Find the power of the pump ($g = 10 \text{ m/s}^2$).
11. State whether kinetic energy is conserved in (a) elastic collisions (b) inelastic collisions.
12. A bullet of mass 0.01 kg moving at 300 m/s gets embedded in a stationary wooden block of mass 2 kg. Find the velocity of the block+bullet system just after impact.
13. A body of mass 3 kg has kinetic energy 24 J. Find its speed.
14. Show that the work done by gravity on a body moving along any path between two points depends only on the vertical height dropped, not on the path taken.

15. A 60 kg person climbs a staircase of height 5 m in 10 s. Find the power developed by the person ($g = 10 \text{ m/s}^2$).
16. A spring with spring constant 100 N/m is stretched from $x = 0.1 \text{ m}$ to $x = 0.2 \text{ m}$. Find the work done in stretching it further.
17. Two bodies of mass 2 kg and 4 kg move towards each other with speeds 3 m/s and 1 m/s respectively and collide perfectly inelastically. Find the velocity after collision.
18. A block slides down a smooth incline of height 5 m starting from rest. Find its speed at the bottom ($g = 10 \text{ m/s}^2$).
19. Distinguish between conservative and non-conservative forces with one example each.
20. A car engine develops 20 kW of power while moving at a constant speed of 20 m/s. Find the driving force delivered.
21. A 2 kg block is projected up a rough incline (angle 30 degrees, $\mu_k = 0.2$) with initial speed 10 m/s. Find how far up the incline it travels before stopping ($g = 10 \text{ m/s}^2$).
22. Explain the difference between average power and instantaneous power, and give the formula for each.

Answer Key

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|--|---|
| 1. $W = 50 \text{ J}$ | 2. $KE = 200 \text{ J}$ |
| 3. $PE = 1 \text{ J}$ | 4. $W(\text{friction}) = -40 \text{ J}$ |
| 5. $P = 250 \text{ W}$ | 6. $P = 20,000 \text{ W}$ (20 kW) |
| 7. $v = 1.5 \text{ m/s}$ | 8. $v_1' = 0$ (first ball stops), $v_2' = 5 \text{ m/s}$ (equal masses exchange velocities) |
| 9. $W = 120 \text{ J}$ | 10. $P = 3920 \text{ W}$ |
| 11. Conceptual - (a) conserved (b) not conserved | 12. $v = 1.49 \text{ m/s}$ (approx) |
| 13. $v = 4 \text{ m/s}$ | 14. Conceptual - see notes on conservative forces |
| 15. $P = 300 \text{ W}$ | 16. $W = (1/2)(100)(0.2^2 - 0.1^2) = 1.5 \text{ J}$ |
| 17. $v = 0.33 \text{ m/s}$ (towards the 2 kg body's original direction, taking signs into account) | 18. $v = 10 \text{ m/s}$ |
| 19. Conceptual - see notes above | 20. $F = 1000 \text{ N}$ |
| 21. s approx 4.46 m | 22. Conceptual - average power = total work/total time; instantaneous power = $F \cdot v$ at that instant |

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.